Table 1: Performance Results

Dataset Size	Algorithm	Average Time (ms)	Average Comparisons/Lookups
1,000	Linear Search	0.058838	479.30
1,000	Recursive Binary Search	0.001911	9.00
1,000	Hash Table Search	0.000417	1.00
10,000	Linear Search	0.897288	4,896.93
10,000	Recursive Binary Search	0.071547	12.20
10,000	Hash Table Search	0.026078	1.00
100,000	Linear Search	12.441707	51,220.87
100,000	Recursive Binary Search	0.021422	15.33
100,000	Hash Table Search	0.001114	1.00
1,000,000	Linear Search	244.479962	521,474.03
1,000,000	Recursive Binary Search	0.009433	18.93
1,000,000	Hash Table Search	0.008276	1.00

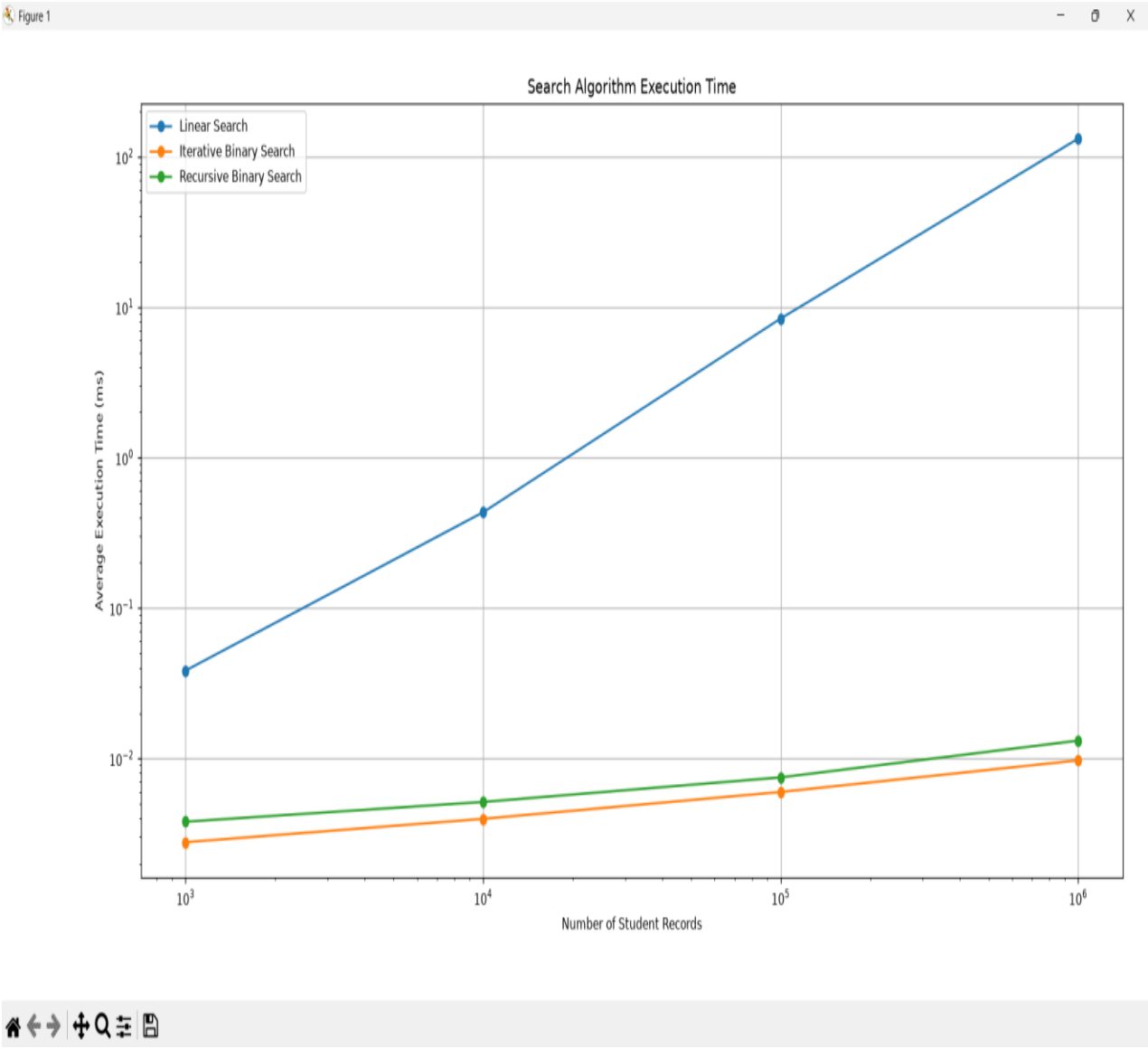
Table 2: Complexity Comparison

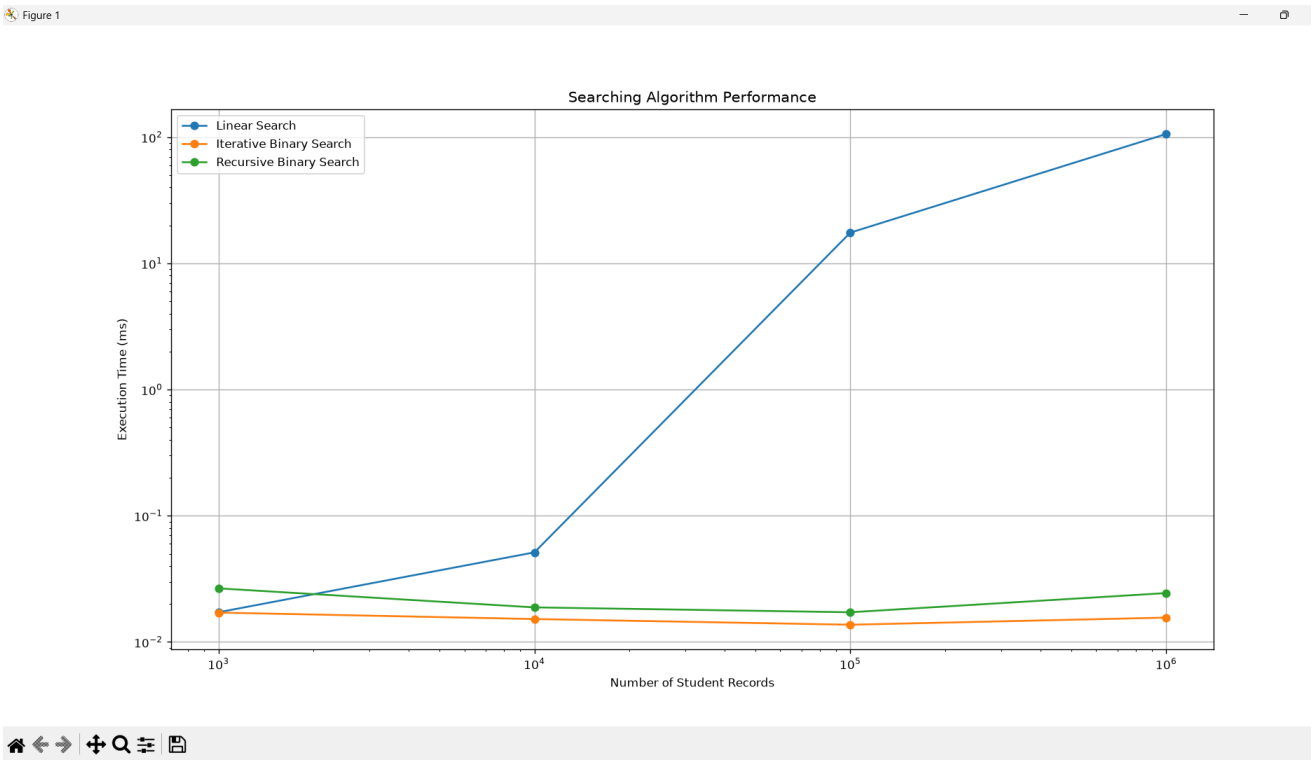
Algorithm	Best Case	Average Case	Worst Case	Auxiliary Space
Linear Search	O(1)	O(n)	O(n)	O(1)
Recursive Binary Search	O(1)	O(log n)	O(log n)	O(log n)
Hash Table Search	O(1)	O(1) expected	O(n)	O(n) for hash table storage

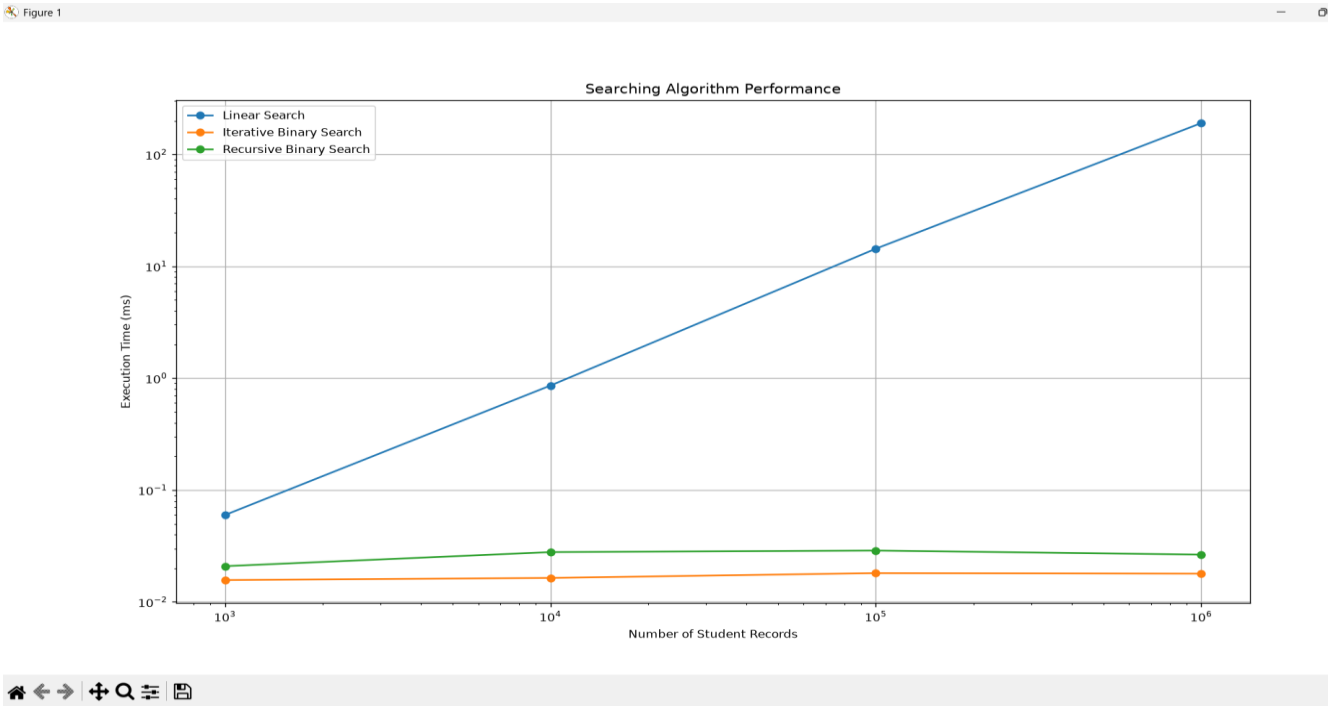
Table 3: Algorithm Comparison

Algorithm	Sorted Input Required	Preprocessing	Advantages	Limitations
Linear Search	No	None	Simple and works on unsorted data	Can require scanning the entire dataset
Recursive Binary Search	Yes	Sort data if initially unsorted	Efficient logarithmic search	Requires sorted data and recursive stack space
Hash Table Search	No	Build and maintain hash table	Expected constant-time lookup	Extra memory and possible collision overhead

Graphs


Figure 1
Search Algorithm Execution Time
Linear Search
Iterative Binary Search
Recursive Binary Search
Average Execution Time (ms)
Number of Student Records





Figures 1 :Seach algorithm Performance

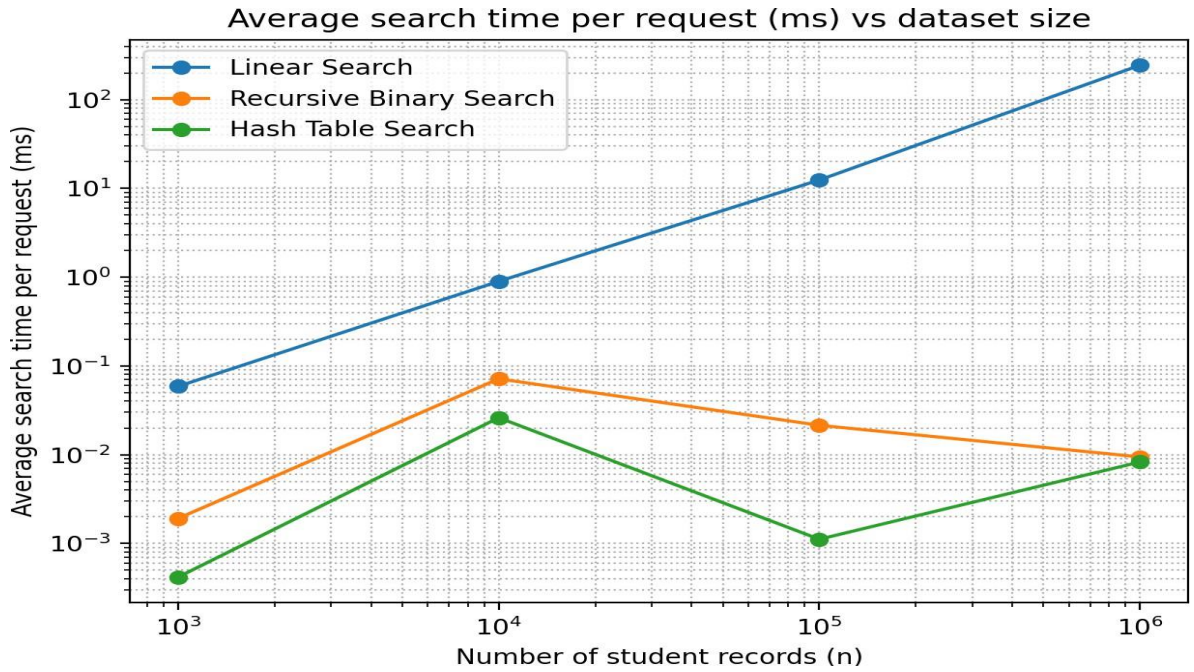


Figure 3. Average per-request search time versus dataset size (log axes).

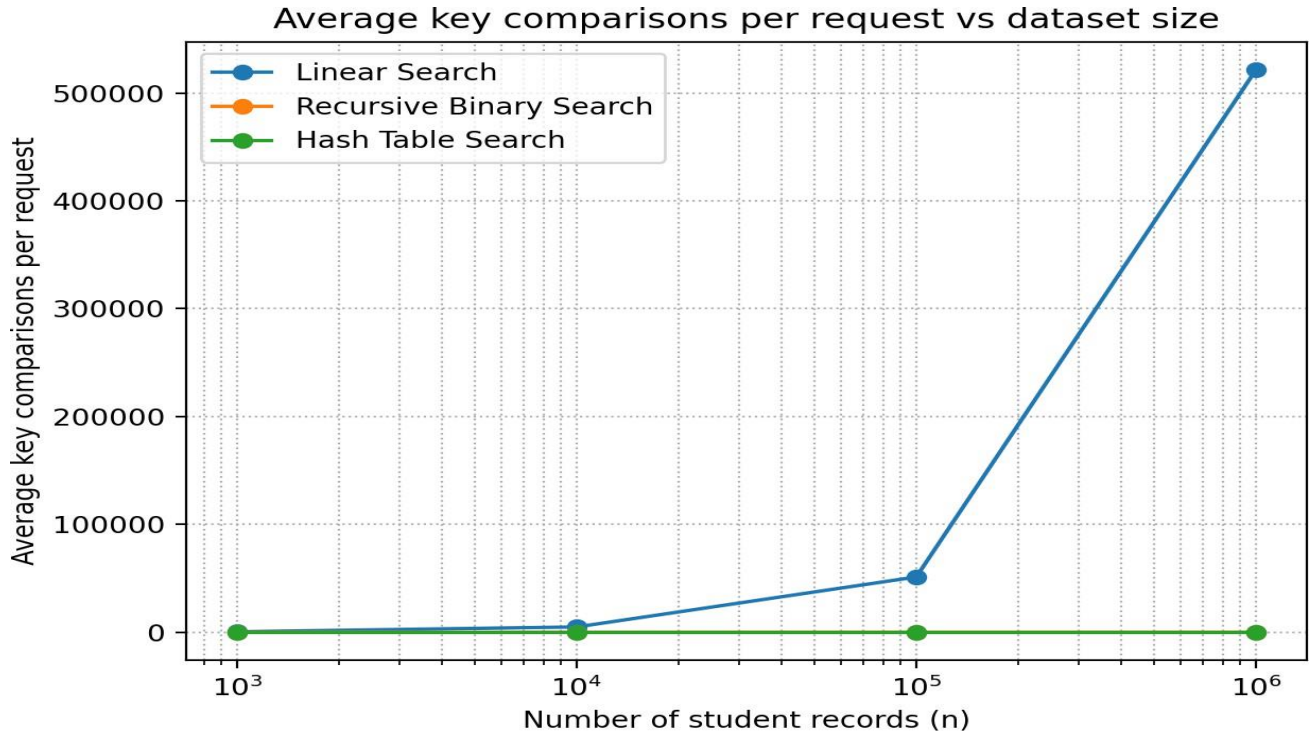


Figure 4. Average key comparisons/lookup operations versus dataset size.